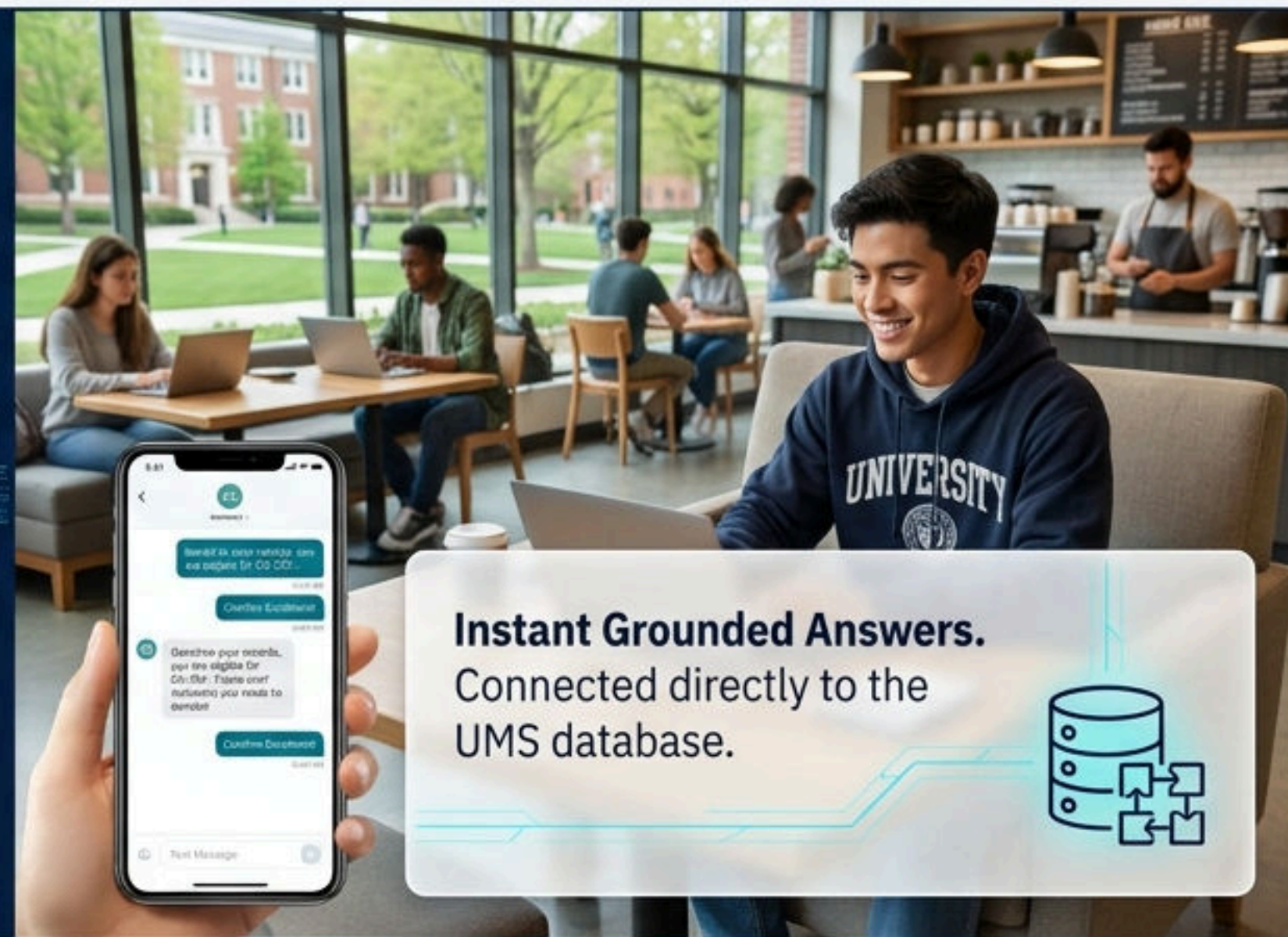
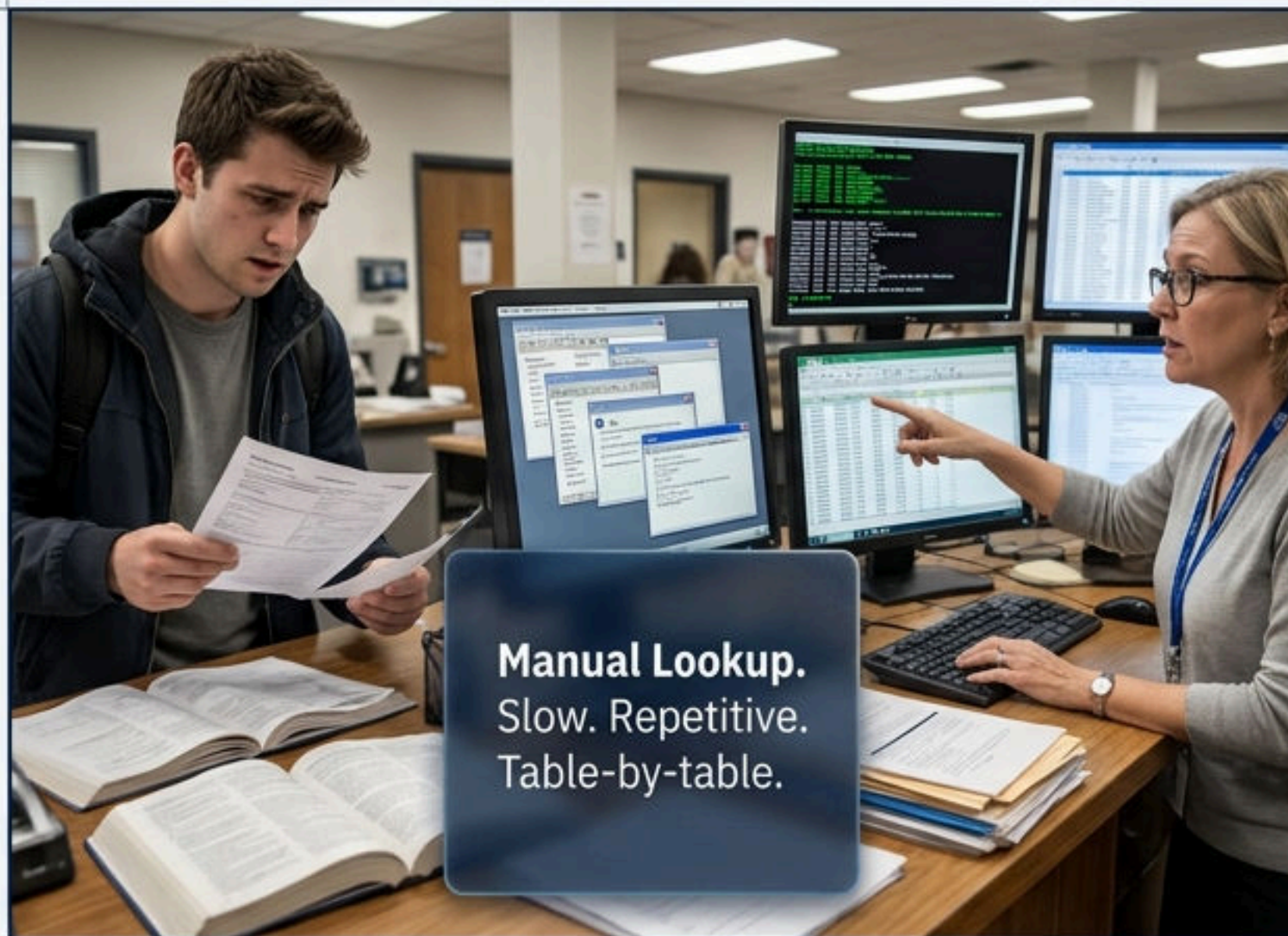


The Registration Bottleneck



Answers pulled directly from the university database—never guessed.



Deterministic eligibility analysis.



Safe execution of enrollments behind a human confirmation gate.

A Dockerized Domain-Specific AI Agent



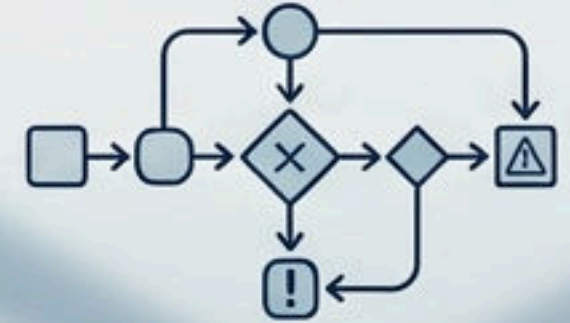
System Role: Decision-support,
not an authoritative system.
Escalates to human staff.

Station 1 (Data)



Maryline Karam (6599)
Tools, database schema,
grounding logic.

Station 2 (Logic)



Aseel Menhem (6651)
LangGraph workflow,
safety gates, LLM prompts.

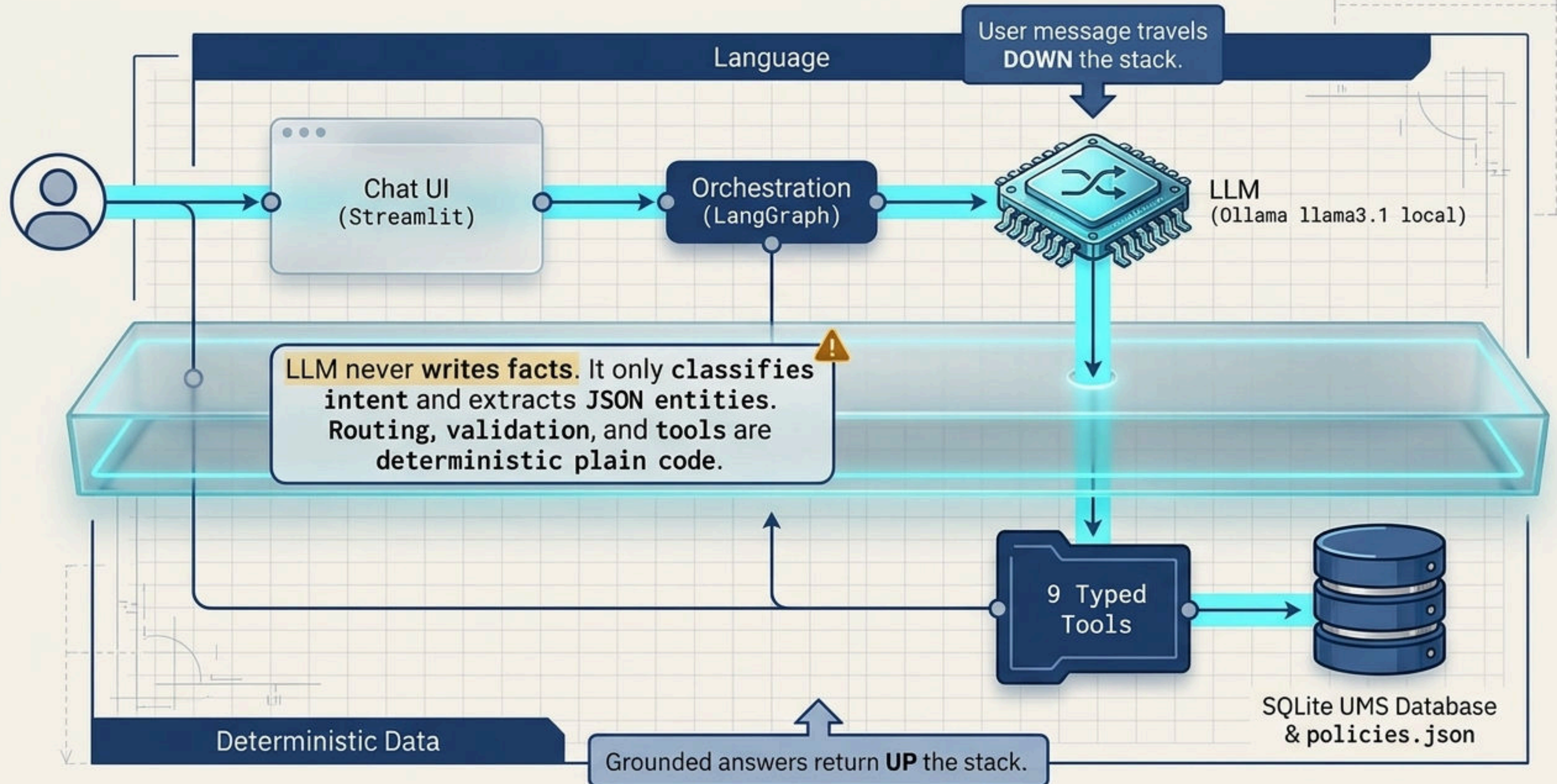
Station 3 (Platform)



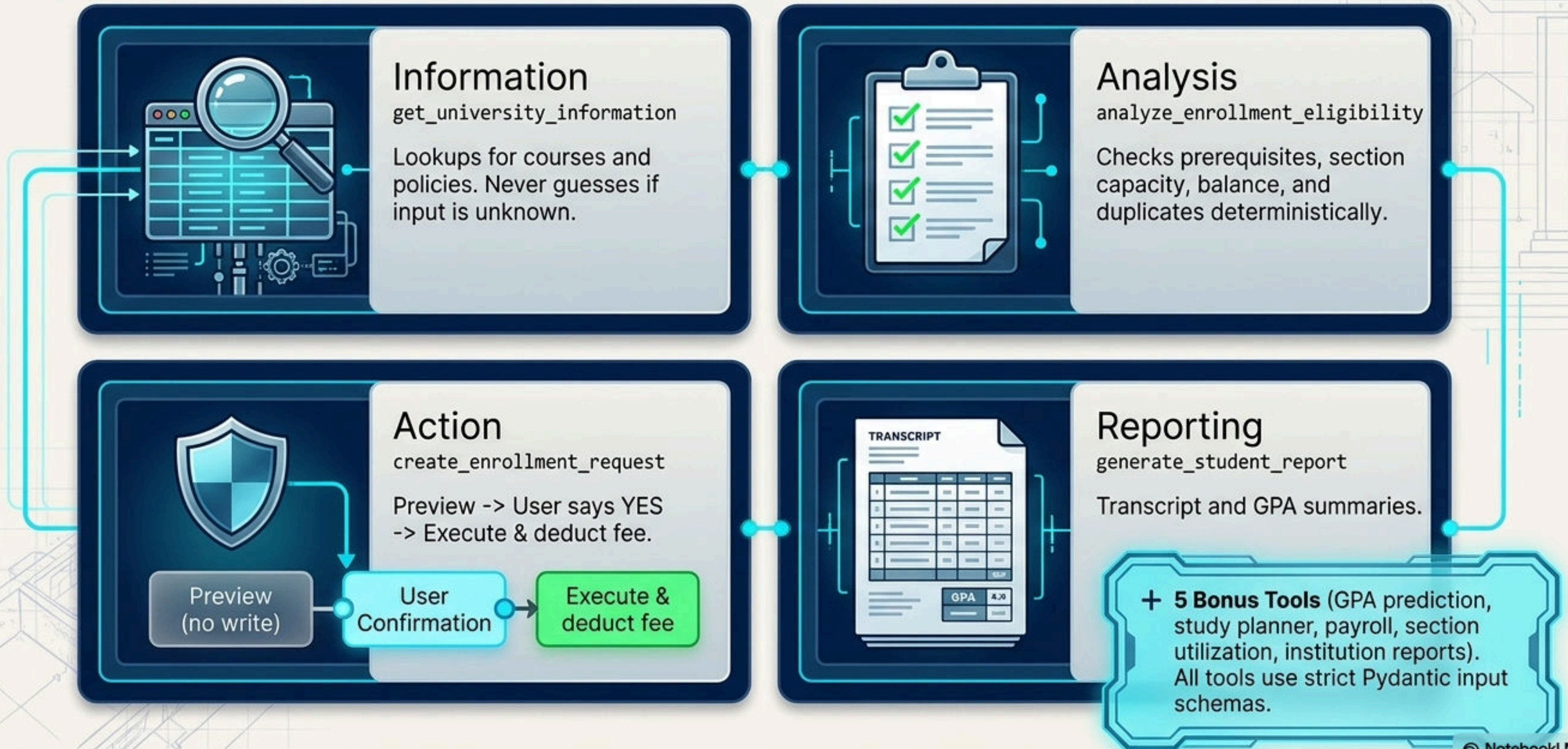
Hana Tfaily (6554)
Memory management, Streamlit UI,
Docker packaging, evaluation.



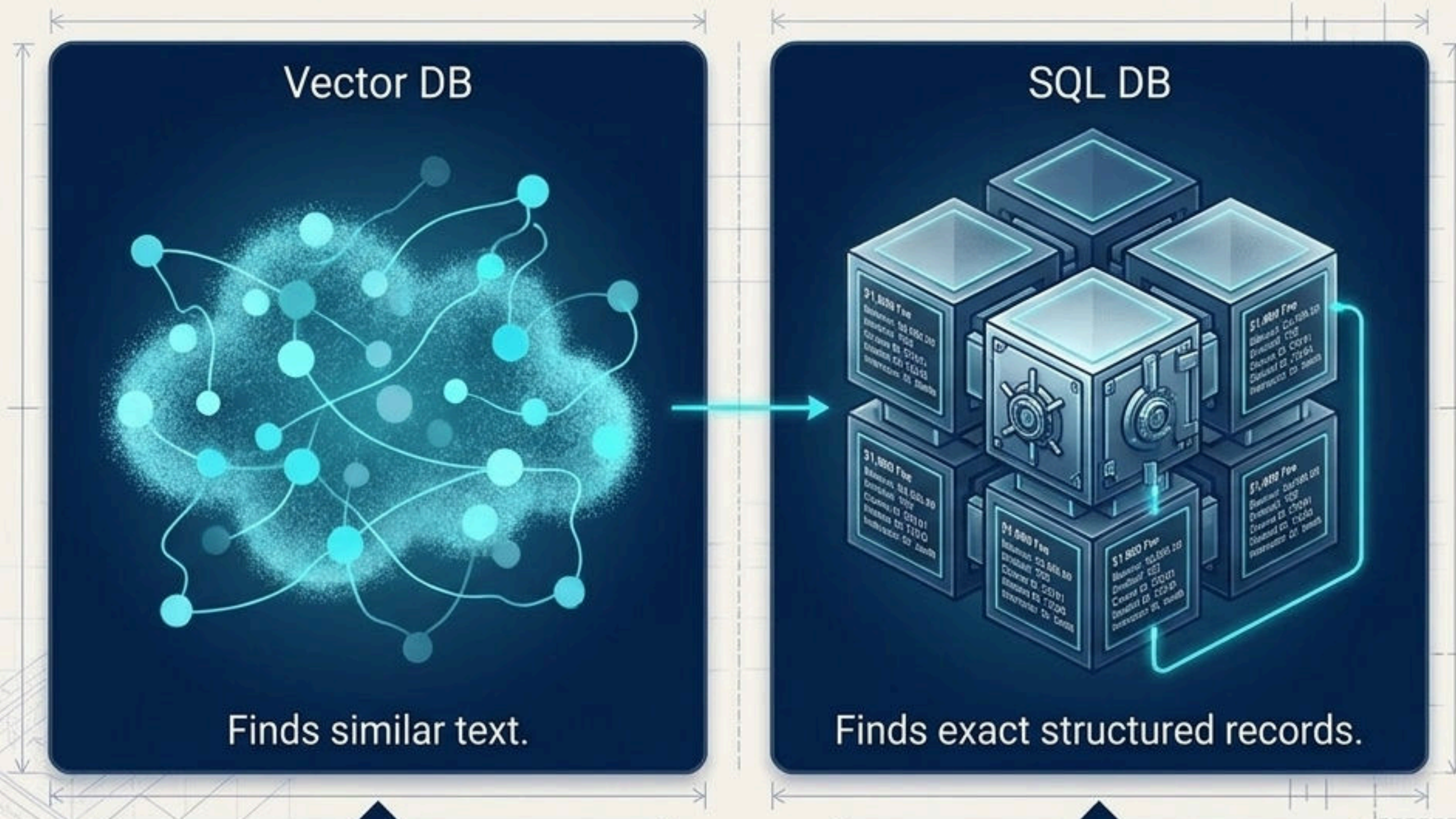
Seven Layers, One Container



Deterministic Tools Over Generative Guesses



Structured Grounding: Why We Rejected RAG



- Domain is small, fully structured, and exactly enumerable.
- Requires exact values for course fees, balances, and enrollment status.
- The same question must always return the exact same mathematical value.

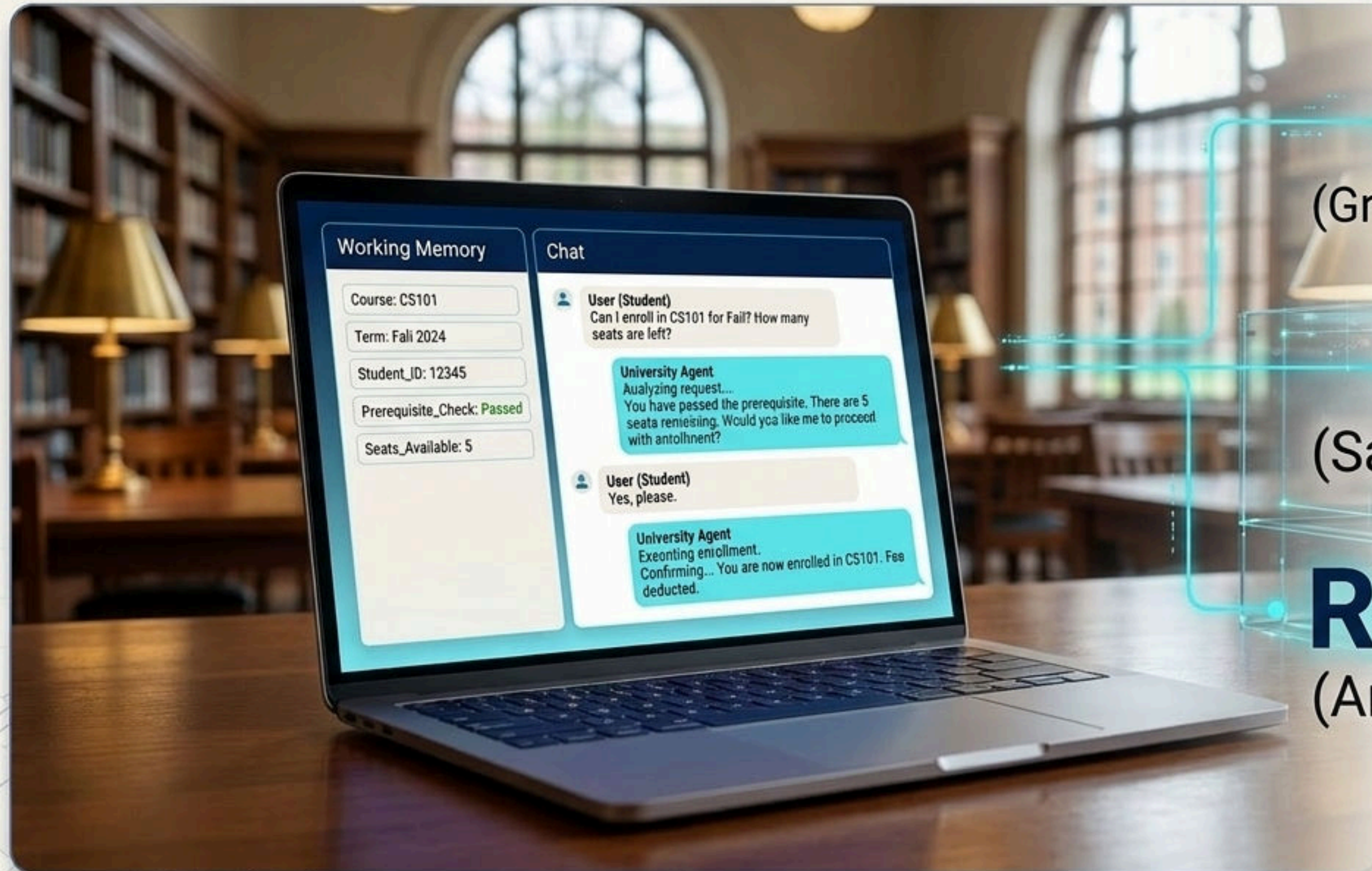


Cannot reliably join, sum, or validate a fee record.



Precise, reproducible, and auditable.

The Agent in Action



Ask.

(Grounded knowledge)

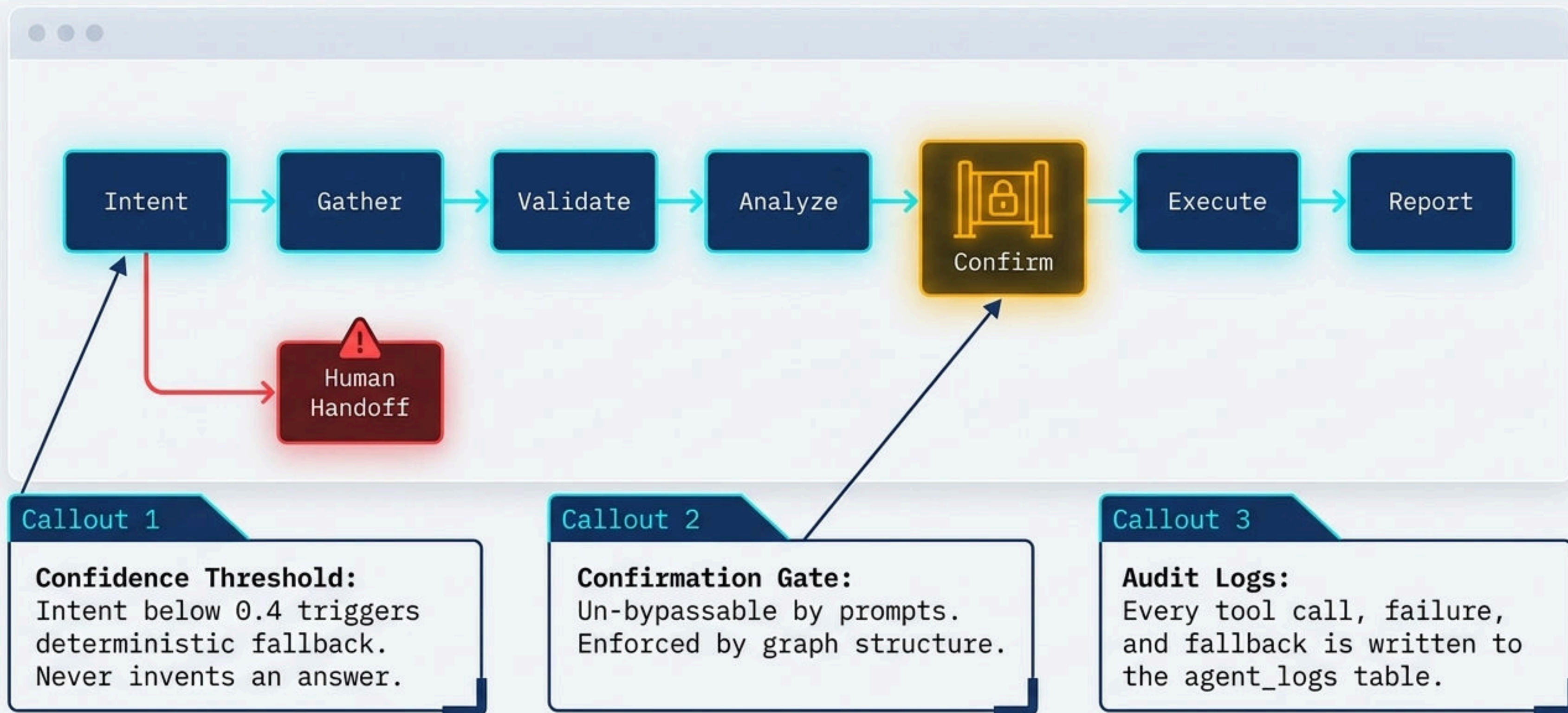
Act.

(Safely)

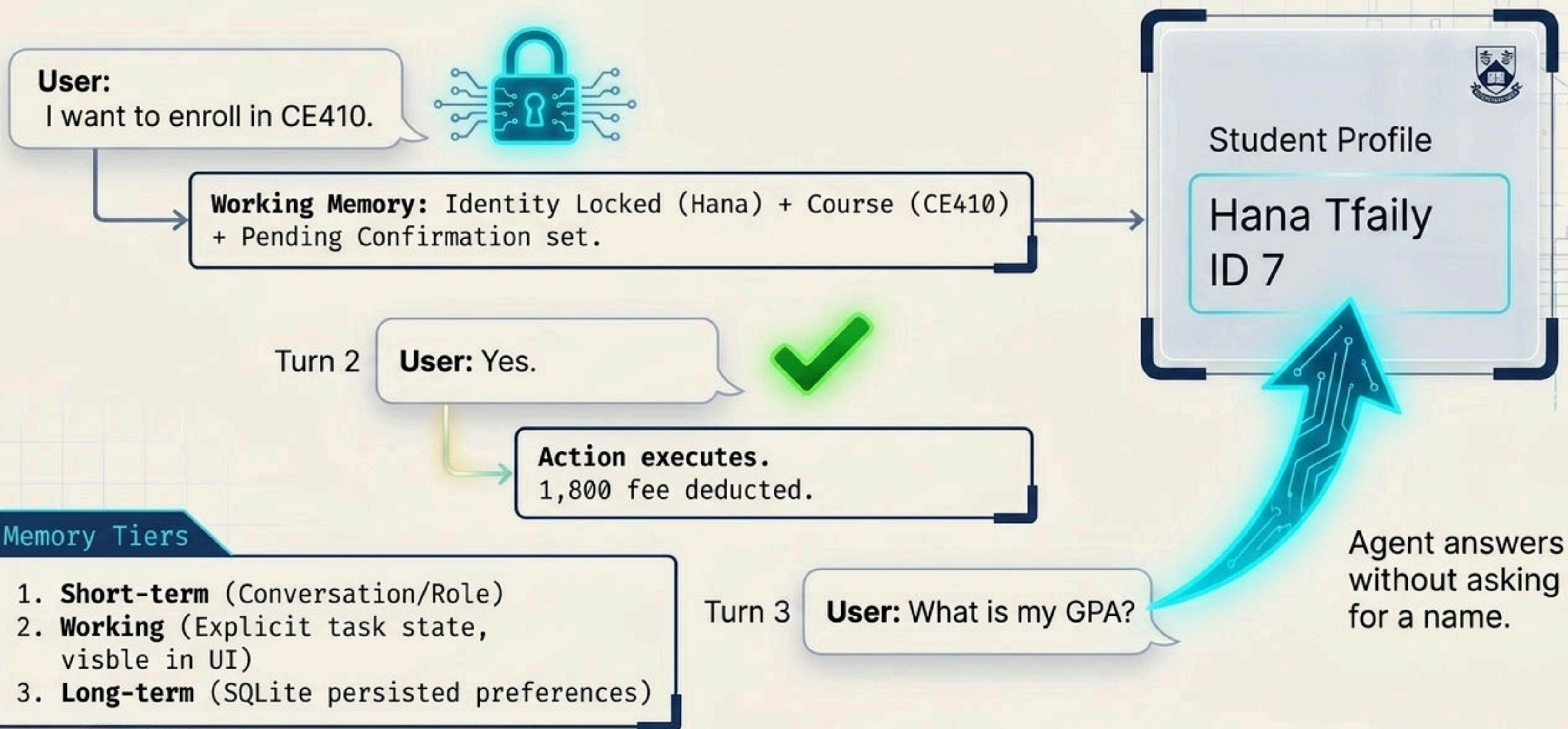
Remember.

(And prove it)

An Explicit State Machine, Not a Free Agent



Contextual Memory Across Three Tiers



Proven by Tests, Reproducible by Docker

```
> docker compose up --build  
Starting local Ollama LLM...  
Starting University Agent...
```

System healthy. Zero API keys required.

Fully offline.
No hard-coded secrets.

94%

Task Completion
(across 35 scripted conversations)

97%

Correct Tool Selection

100%

Fallback Accuracy
(Never guessed)

0

Unsafe
Actions

Confirmation gate never bypassed,
even against prompt injection attacks.

The Transparent Agent: Secure, Automated, Deployed



Academically Robust: Built on a 3NF normalized SQL foundation.



AI Orchestrated: Local, stateless LLM routing layered over deterministic business logic.



Provably Safe: 100% fallback accuracy and zero unauthorized database writes.

All team members (Maryline, Aseel, Hana) are available to discuss any component of the system architecture.